

Player Tracking in Sports — Multi-View Tracking and 3D Reconstruction

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1 Introduction

Vision-based tracking provides a scalable alternative to expensive sports analytics hardware, but basketball’s rapid movements, mutual occlusions, motion blur, and specular floor reflections pose severe challenges. Multi-camera setups can alleviate these occlusions by providing complementary viewpoints, but they require highly robust 2D tracking within each individual camera to ensure consistent cross-view association.

To address these challenges, modern pipelines have largely abandoned classical background subtraction [4] in favor of tracking-by-detection paradigms. Real-time localization is typically driven by single-stage detectors like the YOLO family [6, ?]. To link these spatial detections temporally, SORT [1] established a baseline using Kalman filtering, which was later improved to handle severe occlusions using deep appearance embeddings (DeepSORT [7, ?]) and low-confidence secondary associations (ByteTrack [8]). Finally, multi-camera configurations [4] resolve remaining overlaps by applying epipolar geometry and triangulation to lift 2D sequences into 3D space.

Building on these foundations, we present an end-to-end pipeline for multi-view player tracking and 3D reconstruction, applied to basketball footage from the Sanbapolis facility in Trento, Italy. We systematically evaluate these strategies, demonstrating that a domain-adapted YOLO26 with class-agnostic NMS [5] combined with appearance-based DeepSORT heavily outperforms classical baselines (MOG2, SORT) in reducing identity switches. Finally, we reconstruct 3D trajectories across three calibrated cameras using SVD-based triangulation, utilizing Kalman smoothing to suppress detection noise.

1.1 Dataset and Setup

We evaluate our pipeline on a 21-second basketball sequence (525 frames, 25 fps) captured by three synchronized cameras (cam_13, cam_2, cam_4). Camera intrinsics are provided, while extrinsics are estimated via 2D-3D point correspondences of known court landmarks on the first frame, modeling lens distortion directly within the pose solver. For evaluation, ground-truth bounding boxes are annotated every fifth frame.

2 Detection

2.1 Background Subtraction (MOG2 + CLAHE)

As a baseline, we adopt MOG2 background subtraction [4], in which each pixel’s intensity distribution is modelled as a mixture of K Gaussians:

$$P(x_t) = \sum_{k=1}^K w_{k,t} \mathcal{N}(x_t \mid \mu_{k,t}, \sigma_{k,t}^2). \quad (1)$$

A pixel is classified as foreground if its Mahalanobis distance to the best-matching background component exceeds a fixed threshold. Prior to background modelling, illumination is normalised with Contrast Limited Adaptive Histogram Equalization (CLAHE) [10], and the resulting foreground mask is cleaned with morphological opening and closing; connected components smaller than a minimum area threshold are discarded. Alternative background subtractors (KNN, LSBP, GSOC) were evaluated but offered no improvement over MOG2 while incurring higher computational cost.

Despite these refinements, the method exhibits three recurring failure modes: floor specularities produce shadows whose hue values fall outside the range that MOG2’s built-in shadow detector handles; players who are briefly stationary are indistinguishable from background; and moving spectators generate persistent false positives. Because the optimal morphological kernel sizes and detection thresholds differ across cameras, the method also lacks cross-camera generalization.

2.2 Pre-Trained YOLOv11

To overcome these limitations, we apply a YOLOv11 model [6, 2] pre-trained on COCO, using a confidence threshold of 0.4, an NMS IoU threshold of 0.45, and an inference resolution of 640 px. Detection quality improves substantially, with fewer false positives and better spatial localisation. However, the pre-trained model has two domain gaps: the ball is absent from COCO’s person-centric classes, and players and referees are both labelled “person”, making identity distinction impossible.

2.3 Fine-Tuned YOLOv26 with Multi-Pass Inference

To close both gaps, we fine-tune the YOLOv26 backbone on a Sanbapolis-specific annotated set. The fine-tuned model uses identity-encoding class labels (one class per player identity, e.g. *Red_11*) and a dedicated *Ball* class.

Because at 640 px the convolutional backbone downsamples a 1080p frame by approximately $1.7\times$, a 15 px basketball occupies fewer than two feature-map grid cells, which is insufficient for reliable localisation. We therefore run two separate inference passes per frame:



Fig. 1. Sample detection output from fine-tuned YOLOv26 with class-agnostic NMS on cam_13 (three selected frames). Colour-coded bounding boxes indicate player identities; the ball is shown separately.

- **Player pass:** 1280 px resolution, confidence threshold 0.3. A second run at a lower threshold (0.15) retains a low-confidence band that is reserved for the tracker’s secondary association pass.
- **Ball pass:** 1600 px resolution, confidence threshold 0.2. The higher resolution increases the spatial footprint of the ball, while the lower threshold favours recall for this motion-blurred, occasionally occluded object.

The two sets of detections are merged frame by frame into a single output.

Class-Agnostic NMS Because player identity is encoded directly in the class label, the standard per-class NMS may retain two overlapping boxes that predict different identities for the same physical player. We therefore replace it with a class-agnostic variant [5]: all detections are pooled regardless of class, sorted by confidence, and greedily suppressed so that any box whose Intersection over Union with a higher-scoring box exceeds 0.75 is discarded. This collapses duplicate boxes for the same player without affecting the ball. The result of this step is the final detection output fed to the tracker.

3 Tracking

3.1 SORT

As a baseline tracker, we adopt SORT [1]. Each active track maintains a Kalman filter whose state vector encodes bounding-box position and velocity, $\mathbf{x} = [u, v, s, r, \dot{u}, \dot{v}, \dot{s}]^\top$, where (u, v) is the box centre, s its scale (area), and r the fixed aspect ratio. At each frame, all tracks are propagated forward under a constant-velocity model, and the resulting predicted boxes are matched to incoming detections by the Hungarian algorithm on an IoU cost matrix. Pairs with IoU below a gating threshold are forbidden from matching (max IoU distance 0.8); tracks unmatched for more than 60 frames are deleted, and new detections require two consecutive matches before being confirmed ($n_{\text{init}} = 2$). The main limitation of SORT is that it relies exclusively on spatial proximity for re-identification: when two players cross, their Kalman predictions may overlap, and the tracker frequently swaps their identities.

3.2 DeepSORT

Our final tracker extends SORT with a deep appearance model [7]. For each detected bounding box, OSNet [9]—a lightweight re-identification network—extracts a compact embedding from the cropped image region. Each track maintains a gallery of up to 100 recent embeddings; the cosine distance between a new detection’s embedding and a track’s gallery mean serves as the primary association cost.

Matching proceeds in a cascade ordered by track recency: the most recently observed tracks are matched first using the appearance gate (maximum cosine distance 0.2), and IoU serves as a fallback for tracks that are too new or for which the appearance model is uncertain. This ordering prevents a long-occluded track from claiming a detection through a lucky spatial overlap.

We extend the baseline formulation with four robustness mechanisms: (i) **NSA Kalman update**: the measurement noise covariance is scaled by each detection’s confidence score, so high-confidence boxes correct the state filter more strongly than uncertain ones; (ii) **ByteTrack second pass**: confirmed tracks left unmatched by the appearance cascade are IoU-matched against the low-confidence detections retained from the detection stage [8], recovering players during partial occlusions without contaminating the appearance gallery; (iii) **Track resurrection**: when a confirmed track is deleted, its appearance gallery is retained for a short period; a new detection whose embedding is within the cosine-distance threshold restores the original track identity; (iv) **Coasting**: a briefly-unmatched track continues to emit its Kalman-predicted box, provided it remains on screen and does not conflict with an already-matched box, preserving trajectory continuity over short gaps.

After tracking, 2D trajectories are refined by linearly interpolating gaps of up to five consecutive undetected frames, followed by an exponential moving average ($\alpha = 0.5$) to damp frame-to-frame jitter.

3.3 Label Resolution

Because the fine-tuned detector encodes player identity in the class label, a single track may accumulate detections with slightly different class predictions across frames. For each track, a stable label is assigned by majority vote over all detections, weighted by confidence score. When two tracks claim the same identity, the track with the higher cumulative confidence retains its label, and the other falls back to its next-highest-ranked class. The ball is exempt from this procedure, as it is a category rather than a unique identity.

4 3D Reconstruction

4.1 Camera Calibration

Intrinsic parameters (focal length, principal point, and radial distortion coefficients) are provided directly for each camera. Extrinsic parameters are estimated

by solving the Perspective-n-Point (PnP) problem: a set of court landmarks is manually selected on the first frame of each video, and their 2D image coordinates are paired with known 3D positions in the court coordinate system. Lens distortion is modelled within the PnP solver, so the image does not need to be pre-undistorted.

4.2 Rectification

Before triangulation, the tracked 2D bounding boxes are undistorted using the estimated intrinsics, mapping each detection into a common, distortion-free coordinate system. The reference pixel used for geometry is class-aware: for the ball, the bounding-box centre is used; for players, the bottom-centre of the box is used, as it approximates the ground-contact point and allows placement on the court floor plane.

4.3 Triangulation

Detections are grouped by identity label and frame index across the three cameras. Players and the ball are handled by different strategies.

Players are assumed to stand on the court floor ($Z = 0$). For each camera, the reference bottom-centre pixel is back-projected to a 3D ray and intersected with the ground plane; the resulting per-camera ground estimates are combined by median-inlier averaging, so the 3D position can be recovered even when only a single camera observes a player.

The ball is not constrained to the floor. The projection matrices $P_i = K_i[R_i | t_i]$ for all cameras observing the ball in a given frame are stacked into a linear system

$$\mathbf{x}_i \times P_i \mathbf{X} = \mathbf{0}, \quad (2)$$

which is solved for the homogeneous 3D point $\mathbf{X}$ via SVD (Direct Linear Transform, DLT). Views whose back-projected reprojection error exceeds a threshold are discarded before a final solve to suppress the influence of erroneous detections.

4.4 Trajectory Smoothing

The raw triangulated trajectories contain noise from detection jitter, calibration imprecision, and occasional outliers. Per-entity, a Kalman filter with state vector $[\mathbf{p}, \dot{\mathbf{p}}]$ (position and velocity) is run forward over the trajectory, followed by an RTS (Rauch–Tung–Striebel) backward smoother, so each point is corrected using both past and future observations. Measurements whose Mahalanobis distance exceeds a chi-square gate are rejected as outliers, and trajectories are split wherever the gap between consecutive observations exceeds a maximum length to avoid extrapolation over long occlusions.



Fig. 2. Reconstructed 3D player and ball positions projected onto the court plane (top view) after RTS smoothing. Each colour corresponds to a unique player identity.

5 Evaluation Metrics

5.1 Detection Metrics

A predicted box is a True Positive (TP) when its Intersection over Union (IoU) with an unmatched ground-truth box satisfies $\text{IoU} \geq 0.5$; otherwise it is a False Positive (FP), and a ground-truth box with no matching prediction is a False Negative (FN). The standard metrics are:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2TP}{2TP + FP + FN}. \quad (3)$$

Mean IoU reports the average spatial overlap of matched TP pairs, independently of precision and recall.

5.2 Tracking Metrics

IDF1-Based Identity Metrics IDF1 measures how consistently predicted tracks maintain the correct identity. Detections are matched by IoU (≥ 0.5) and the predicted identity is compared with the ground-truth class name. Identity True Positives (IDTP) are correctly matched pairs; Identity False Positives (IDFP) and Identity False Negatives (IDFN) penalise identity confusion and missed detections respectively. ID Precision (IDP), ID Recall (IDR), and their harmonic mean IDF1 follow the same formulas as ordinary Precision, Recall, and F_1 .

HOTA HOTA [3] decomposes tracking quality into independent detection and association components evaluated at multiple IoU thresholds $\alpha \in \{0.05, 0.10, \dots, 0.95\}$. Detection accuracy is:

$$\text{DetA}(\alpha) = \frac{TP(\alpha)}{TP(\alpha) + FP(\alpha) + FN(\alpha)}. \quad (4)$$

Association accuracy quantifies how consistently matched detection pairs are linked into the same track across frames:

$$\text{AssA}(\alpha) = \frac{1}{|TP(\alpha)|} \sum_{(c, \hat{c}) \in TP(\alpha)} \frac{|TPA(c, \hat{c})|}{|TPA(c, \hat{c})| + |FPA(c, \hat{c})| + |FNA(c, \hat{c})|}, \quad (5)$$

where TPA , FPA , FNA count per-pair association correct, wrong, and missed frames. HOTA at a single threshold is their geometric mean:

$$\text{HOTA}(\alpha) = \sqrt{\text{DetA}(\alpha) \times \text{AssA}(\alpha)}, \quad (6)$$

and the final score is the uniform average over all thresholds. LocA reports the mean IoU of all matched pairs, summarising spatial localisation independently of association.

5.3 3D Geometry Metrics

Reprojection Error Each triangulated 3D point is projected back into the image plane of every camera via $\hat{\mathbf{p}} = K[R \mid t]\mathbf{P}$, and the Euclidean distance to the 2D detection that served as the triangulation input is computed: $e = \|\hat{\mathbf{p}} - \mathbf{p}_{\text{pred}}\|_2$. We report the mean, median, RMSE, and the fraction of matches below 5, 10, and 20 pixels.

Trajectory Error Per-identity predicted pixel trajectories (obtained by projecting the 3D track into each camera) are compared against ground-truth 2D annotated trajectories at every shared annotated frame. We report:

- **ADE** (Average Displacement Error): mean Euclidean distance over all shared frames.
- **FDE** (Final Displacement Error): Euclidean distance at the last shared frame; captures long-term drift.
- **MTE** (Median Trajectory Error): median per-frame displacement, robust to outlier frames.
- **Fragments**: number of coverage gaps in the predicted trajectory relative to the ground-truth span.

6 Results and Discussion

6.1 Detection

Table 1 reports detection performance on the annotated frames.

MOG2 achieves the lowest F_1 on all cameras, confirming that the assumption of a static background is violated by moving spectators and specular reflections. The pre-trained YOLOv11 baseline generalises far better, but recall remains limited because the ball is absent from its label set and some player detections are missed in crowded regions. The fine-tuned model with class-agnostic NMS achieves the best F_1 on cam_4 (0.870) and competitive scores on the other views, with a notably higher precision than the pre-trained model on cam_4 (0.923 vs. 0.927) and substantially better recall on cam_13 (0.732 vs. 0.602), demonstrating the value of domain adaptation.

Table 1. Detection results across cameras (IoU threshold = 0.5).

Method	Camera	Precision	Recall	F_1	Mean IoU
MOG2	cam_13	0.471	0.262	0.337	0.651
	cam_2	0.345	0.408	0.374	0.663
	cam_4	0.328	0.293	0.309	0.700
YOLO pre-trained	cam_13	0.834	0.602	0.699	0.748
	cam_2	0.819	0.781	0.800	0.785
	cam_4	0.927	0.807	0.862	0.808
YOLO fine-tuned	cam_13	0.872	0.732	0.796	0.758
	cam_2	0.835	0.761	0.796	0.757
	cam_4	0.923	0.822	0.870	0.796

Table 2. Identity tracking metrics (IoU threshold = 0.5).

Tracker	Camera	IDP	IDR	IDF1
SORT	cam_13	0.660	0.514	0.590
	cam_2	0.743	0.607	0.668
	cam_4	0.768	0.605	0.679
DeepSORT	cam_13	0.588	0.533	0.569
	cam_2	0.617	0.603	0.610
	cam_4	0.670	0.605	0.638

6.2 Tracking

Tables 2 and 3 report identity and HOTA scores.

On IDF1, SORT outperforms DeepSORT on this short clip, while HOTA scores are close between the two across all cameras. The identity-in-class labels from the fine-tuned detector already encode player identity at detection time, which benefits SORT’s simple IoU matching. DeepSORT’s advantage lies in its greater resilience to prolonged occlusions through appearance-based re-identification; however, over only 525 frames, such events are relatively infrequent, and the increased track count from coasting and resurrection slightly raises IDFP. The association component (AssA) is broadly similar for both trackers, confirming that the main tracking challenge on this clip is detection recall rather than association quality.

6.3 3D Reconstruction

Tables 4 and 5 report geometry quality.

Reprojection error is substantially lower for cam_13 (median 28.9px) than for the other two views (median 94.5 and 50.0px), reflecting the relative quality of the manually selected calibration correspondences. The large RMSE values for cam_2 and cam_4 indicate the presence of outlier frames where the triangulated

Table 3. HOTA tracking metrics.

Tracker	Camera	HOTA	DetA	AssA	LocA
SORT	cam_13	0.438	0.484	0.398	0.797
	cam_2	0.517	0.515	0.522	0.794
	cam_4	0.542	0.575	0.514	0.829
DeepSORT	cam_13	0.434	0.469	0.405	0.769
	cam_2	0.480	0.453	0.511	0.761
	cam_4	0.521	0.553	0.495	0.798

Table 4. Reprojection error of triangulated 3D points.

Camera	Mean (px)	Median (px)	RMSE (px)	Acc@5px	Acc@10px	Acc@20px
cam_13	51.3	28.9	87.7	0.135	0.221	0.349
cam_2	140.0	94.5	278.7	0.028	0.047	0.094
cam_4	135.0	50.0	296.2	0.168	0.214	0.295

point reprojects far from the observed detection, likely due to a poor extrinsic estimate or a detection on a mismatched identity.

The trajectory errors (ADE/FDE/MTE in the hundreds of pixels) are large across all views, revealing that the primary bottleneck is not the tracking or triangulation algorithms but calibration accuracy: a systematic offset in the estimated camera pose propagates directly into a constant positional bias in the reconstructed trajectories. The fragment count is low (2–5 per camera), indicating that the reconstructed tracks cover the annotated span with few coverage gaps.

6.4 Limitations and Failure Cases

Several systematic failure modes remain. Prolonged occlusions (more than ~ 2.4 s at 25 fps) exhaust the tracker’s maximum age, causing track deletions and subsequent identity re-assignment even with appearance-based resurrection. Ball detection is unreliable under severe motion blur or when the ball passes directly under a player, reducing its visible area below the confidence threshold. Partial court coverage by cam_2 and cam_4 means that a player near the court boundary may be visible in only one or two views, reducing the robustness of triangulation. Finally, the manually determined extrinsic calibration introduces biases that directly inflate the reprojection and trajectory errors reported above; an automatic or online calibration procedure would be necessary for deployment.

7 Conclusion

We presented an end-to-end pipeline for multi-view basketball player and ball tracking and 3D reconstruction, evaluated on a three-camera recording from the

Table 5. Trajectory error of projected 3D tracks vs. ground-truth 2D annotations.

Camera	ADE (px)	FDE (px)	MTE (px)	Fragments
cam_13	976.8	849.8	1047.5	2
cam_2	1066.0	1279.1	1181.4	5
cam_4	1838.2	1689.1	1899.9	5

Sanbapolis facility. Across the detection stage, domain fine-tuning of YOLOv26 with multi-pass inference and class-agnostic NMS consistently outperforms both MOG2 and the COCO pre-trained baseline, demonstrating that even a modest annotated domain-specific set is sufficient to bridge the gap between general and sports-specific detection. For tracking, the extended DeepSORT formulation—incorporating OSNet appearance embeddings, ByteTrack low-confidence recovery, and track resurrection—improves identity continuity over longer occlusions, though on this short clip the SORT baseline remains competitive thanks to the identity-rich class labels produced by the detector. 3D reconstruction via SVD-based triangulation followed by RTS-smoothed Kalman filtering yields continuous per-player and ball trajectories; the dominant source of residual error is manual extrinsic calibration bias rather than the triangulation or smoothing procedure.

Future directions include replacing the manual calibration step with an automatic or online method driven by tracked court line features, extending the evaluation to full-game sequences that expose the tracker’s long-term identity maintenance behaviour, and investigating learned triangulation approaches that jointly optimise detection confidence and geometric consistency across views.

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